Bridging the Explainability Gap In Digital Marketing: A Structural Equation Model for Transparent Media Mix Modeling

Anonymous Author(s)

Abstract

The rapid evolution of digital marketing has caused a high level of competition, compelling marketing companies to rely heavily on Media Mix Modeling (MMM) to optimize their budgets. However, traditional MMM models often lack the causal transparency required for strategic decision-making. This study bridges this explainability gap by analyzing the interdependent causal mechanisms between marketing channels and business performance, while measuring how managerial response to environmental volatility affects revenue. Using Structural Equation Modeling (SEM) on 135 weeks of proprietary US market data from a leading global consumer electronics manufacturer, the theory-driven structural model ($R^2 = 0.82$) reveals the causal relations among factors in different levels of the marketing funnel that traditional predictive methods fail to identify. The model demonstrates high robustness and avoids over-fitting, validated structurally via strong fit indices (CFI = 0.98, RMSEA = 0.03) and predictively through a chronological walk-forward validation, yielding an out-of-sample Mean Absolute Percentage Error (MAPE) of 8.81%, offering a transparent, causal approach to evaluating marketing interventions and optimizing strategic resource recommendations.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Modeling methodologies.*

Keywords

Media Mix Modeling, Structural Equation Modeling, Recommendation Systems, Causal Inference, Explainable Analytics, Resource Allocation

ACM Reference Format:

Anonymous Author(s). 2026. Bridging the Explainability Gap In Digital Marketing: A Structural Equation Model for Transparent Media Mix Modeling. In *Proceedings of 20th ACM Conference on Recommender Systems (RecSys'26).* ACM, New York, NY, USA, 8 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Background of the Study

Measuring the true causal effects of digital advertising remains a challenge despite the availability of granular data [6]. Marketing companies often use algorithmic models to estimate return on investment (ROI), but these lack explainability—they identify statistical correlations rather than underlying causal structures. Recent studies and experiments indicate that models based solely on observational data consistently overstate the effectiveness of lower-funnel, high-intent channels, while simultaneously underestimating the demand-generation potential of upper-funnel media [6]. This divergence, often referred to as channel influence bias [9], demands a transition from basic attribution methods to robust causal inference techniques to avoid the systematic misallocation of resources [2].

Media Mix Modeling (MMM) functions essentially as a macroscopic recommender system. Rather than predicting the probability of an individual user interacting with a specific item, MMM forecasts the aggregate acceptance of digital interventions across diverse media platforms. This approach broadens the scope of personalization, moving beyond individual users to include larger groups defined by shared behaviors, search patterns, or online communities [9, 12].

Marketers employ the marketing and sales funnel to map user engagement across digital platforms and monitor the progression toward a purchase [3]. The upper-funnel segment is directed toward potential users who lack familiarity with the brand or its products. During this phase, advertisements are presented based on general contextual relevance to create widespread brand recognition and collect preliminary behavioral data. Conversely, the lower-funnel segment concentrates on qualified leads demonstrating specific, high-intent behaviors, thus facilitating direct revenue attribution, given their elevated probability of completing a purchase.

Traditionally, media mix models (MMMs) have relied heavily on multiple linear or ridge regression to predict revenue based on past media spending [9]. However, these single-equation models oversimplify the sequential decision-making process. By treating all media channels as separate, parallel variables, they don't fully capture the hierarchical relationships between upper-funnel awareness and lower-funnel intent [9]. Although advanced predictive machine learning models can reduce errors outside the sample, they inherently obscure these structural pathways, functioning as black box systems that offer limited strategic transparency. Therefore, predictive accuracy alone is insufficient for effective resource allocation when the underlying causal relationships are not clearly defined [2].

To address the limitations in explainability of opaque systems, Structural Equation Modeling (SEM) has been used, as it offers a mathematically sound approach for representing intricate, underlying relationships that are not adequately addressed by conventional analytical methods. SEM has been successfully utilized by researchers in various computational fields to forecast technology adoption [7, 8], assess marketplace loyalty [11], and elucidate the processes underlying initial user trust [1]. Furthermore, SEM has

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RecSys'26, Minneapolis, MN, USA

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

demonstrated considerable utility within the user modeling community, particularly in quantifying the effectiveness of persuasive strategies designed for distinct user groups [12].

We propose using Covariance-Based SEM to examine causal relationships in digital marketing. Analyzing 135 weeks of proprietary U.S. market data from a leading global consumer electronics manufacturer, the model suggests and tests a multi-step causal process. This process shows how upper-funnel media spending creates hidden demand, which is then captured by lower-funnel channels [5]. In addition, the study introduces and measures a new concept, *Budget Allocation Instability*, to assess the hidden costs of decision-making problems. It suggests that reactive budgeting, which responds to market changes, leads to inefficient outcomes [14]. By clarifying these interactions, the study offers a clear recommendation system for strategic media planning.

To systematically uncover these causal mechanisms, this research tests four structural hypotheses grounded in consumer economics and organizational behavior:

Managerial Behavioral Response Dynamics (H1): Decision-makers often make constrained, reactive decisions during market uncertainty [14]. It is hypothesized that environmental stressors (e.g., market disruption, revenue volatility) drive budget allocation instability, and that this reactive shifting of resources fails to generate long-term revenue improvements.

Direct Revenue Attribution and Environmental Controls (H2): Isolating advertising causality requires controlling for external baselines[2]. It is hypothesized that high-intent, lower-funnel channels (e.g., Brand Search, LinkedIn, Reddit) directly drive revenue, whereas generic Non-Brand Search yields lower direct returns. Furthermore, it is hypothesized that macro-environmental factors (e.g., competitor spending, seasonality, holidays, macroeconomic inflation) dictate baseline sales volumes.

Mediated Funnel Mechanics and Consumer Economics (H3): Consumers engage with multiple touchpoints before purchasing [5]. Users navigate multiple digital touchpoints prior to conversion. It is hypothesized that upper-funnel media (e.g., Meta, Display) operate their influence indirectly by generating the initial awareness that stimulates downstream, high-intent search behaviors captured by lower-funnel channels.

Strategic Resource Allocation Interdependencies (H4): Media spend allocation involves complex cross-channel effects [9]. It is hypothesized that strategic bundling (e.g., grouping social, video, audio), and platform substitution effects, particularly between hybrid video platforms and traditional search ecosystems.

This study aims to transform macroscopic resource allocation from a solely predictive, opaque process into a causally-informed recommendation engine[1] by empirically testing these four assumptions using Covariance-Based Structural Equation Modelling (CB-SEM). The resulting structural of this study is beneficially important for marketing organizations. To stop automated algorithmic misallocation, marketing analysts need to be able to measure hidden sequential mediation (such upper-funnel awareness driving search intent) and figure out the structural managerial decision friction. Moreover, the identified causal pathways function as a diagnostic triage instrument, allowing marketing teams to prioritize ambiguous or high-leverage channels for focused Matched Market geo-experiments before the implementation of comprehensive Bayesian Media Mix Models.

The remainder of this research is organized as follows: Section 2 reviews related work concerning algorithmic attribution and the explainability gap in macroscopic recommender systems. Section 3 details the empirical methodology, including the formal specification, out-of-sample validation protocol, and the CB-SEM framework. Section 4 presents the results of the structural estimation and hypothesis testing, alongside a benchmark comparison against traditional baseline regressions. Section 5 discusses the broader theoretical and practical implications of the findings. Finally, Section 6 concludes the study and outlines directions for future experimental integration.

2 Related Works

This section is about the theoretical and recent writing on macroscopic resource allocation. In particular, it looks at how advertising attribution has changed to include causal inference, how traditional Media Mix Modelling (MMM) falls short when used as an automated recommendation engine, and how Structural Equation Modelling (SEM) can be used to map complex behavioural systems. This section identifies the essential explainability and methodological limitations that the proposed structural framework addresses.

Causal Inference and Incrementality in Advertising

The shift from deterministic attribution to causal inference represents a central area of focus within modern marketing analytics. Conventional multi-touch attribution models frequently exhibit channel influence bias, thereby resulting in the inefficient allocation of organizational resources [9]. Recent studies highlight the importance of incrementality testing for the validation of observational models. As an illustration, large-scale blackout experiments reveal that observational attribution often overstates the direct effectiveness of high-intent channels while underestimating the demand-generation potential of upper-funnel media [2]. Likewise, extensive field experiments expose substantial divergences between standard algorithmic predictions and actual causal lift [6]. As a result, these findings highlight the limitations of purely predictive algorithms in situations characterized by complex user experiences.

Structural Equation Modeling for User Behavior

Researchers are increasingly utilizing structural equation modelling (SEM) to examine complex relationships and hidden variables. SEM has been widely used in the field of information systems to identify relationships that single-equation models can't easily capture. In user modeling, SEM has been successfully used to measure how well persuasive strategies work for different user groups [12], demonstrating its value in segmenting and predicting group behaviors. Additionally, structural frameworks have been used to understand the hidden factors that build initial user trust in data-driven systems [1]. In addition, SEM offers a strong mathematical basis for identifying factors that influence adoption in complex environments, such as peer-to-peer energy sharing [7] and mass

[1]To support open science and reproducibility, the anonymized enterprise dataset and Python estimation codebase are available for peer review at: https://anonymous.4open.science/r/bridging-explainability-mmm-E220

customization technologies [8], going beyond just user modeling.

Advancements in Media Mix Modeling

Media mix modeling (MMM) frameworks are being developed to address the effects of various channels and macro-environmental factors. Recent research have shifted towards "out-of-the-box" AI MMMs that automate complex hyperparameter optimization and feature engineering [4]. However, even while predictive machine learning techniques demonstrate proficiency in predicting out-of-sample variance, they often obscure underlying structural dependencies, functioning as black-box systems. Such systems operate as opaque engines that struggle to distinguish between correlation and true causality without extensive manual intervention and domain-specific priors [4]. Our research expands upon these existing studies by using Covariance-Based SEM not solely as a statistical evaluation tool, but as a transparent, macroscopic recommendation engine. By clearly describing the sequential flow of user intent, the proposed model mitigates channel influence bias and provides practical, causal resource allocation strategies that pure predictive correlation cannot consistently produce.

Research Gap and Present Contribution

Although existing research explores the predictive capabilities and automation of AI-driven MMMs [4] and the application of SEM in isolated user modeling contexts [12], a critical gap remains at their intersection. Prior studies primarily treat digital marketing channels as independent variables or rely on opaque machine learning algorithms that prioritize out-of-sample forecasting over causal explainability [4, 9].

Our study diverges from these traditional approaches in three distinct ways:

Theoretical Integration: Bridging the gap between cohort-based user modelling and aggregate marketing analytics by conceptualizing MMM as a macroscopic recommender system.

Causal Transparency: Instead of using opaque methods, CB-SEM, a mathematically transparent "white-box" model approach, is used to provide causal transparency. This method makes it easier to measure indirect, sequential mediation effects that are frequently missed by traditional predictive regressions, like the impact of upper-funnel awareness on lower-funnel intent [5].

Human-in-the-Loop Dynamics: This study presents and empirically assesses "managerial decision friction" (Budget Allocation Instability) [14], a behavioral factor infrequently considered in resource allocation models that rely solely on algorithms.

By addressing these gaps, this research goes beyond superficial predictive correlations, providing a practical, causally grounded framework for resource recommendation.

3 Methodology

In this section, we present the data source, model constructs and variables, the formal model specification and its composition based on the four hypotheses listed in the Introduction.

Data Collection and Research Variables

To validate the structural hypotheses, this study examines 135 consecutive weeks of U.S. retail data sourced from a leading multinational consumer electronics brand. The data were aggregated on a weekly basis to improve statistical robustness and to reduce the high-frequency volatility characteristic of daily digital advertising metrics. Within this study, "revenue" signifies the total return on media investment derived from direct retail sales, whereas "spend" refers to the financial resources dedicated to executing interventions across designated digital platforms.

To show the hierarchical structure of the marketing funnel and assess cross-channel relationships, Table 1 categorizes the observed variables into separate theoretical categories, each reflecting user intent. These categories operationalize the macroscopic recommender framework. Specifically, "Upper-Funnel Media" targets broad cohorts with low initial brand affinity to drive awareness, whereas "Lower-Funnel Media" captures users exhibiting specific, downstream conversion behaviors. "Hybrid Media" represents dual-intent platforms that serve both exploratory and direct-response functions. To isolate true advertising incrementality from baseline market fluctuations, macroeconomic and volatility metrics are formulated as "Environmental Controls and Stressors" [2]. Finally, to capture the behavioral aspect of resource allocation, "Managerial Dynamics" introduces a calculated metric of Budget Allocation Instability, utilized to measure decision friction over time [14].

Formal Model Specification

Complex systemic behaviors, such as mass customization in manufacturing [8], require flexible latent variable path analysis. Mirroring these methodological approaches, a Covariance-Based Structural Equation Model (CB-SEM) was constructed to map the marketing funnel's causal relationships. The model was estimated via Maximum Likelihood (ML) using the semopy library in Python. To satisfy the distributional assumptions required for ML estimation and to enable direct comparison of path coefficients, all continuous variables were standardized prior to analysis.

Rather than relying on isolated, single-equation regressions, the structural model is conceptualized as a system of simultaneous equations. This approach allows downstream media spend and revenue to be modeled as functions of both interconnected internal variables (representing mediated funnel mechanics) and external variables (representing environmental stressors). Consequently, the model isolates direct, indirect, and total effects across the sequential user experience. To meet all the hypotheses required for parametric estimation and to enable the direct comparison of structural path coefficients, all continuous variables were standardized prior to analysis.

Table 1: Measurement Constructs and Indicators Categorized by User Intent

Construct Category	Variables / Channel Inclusions
Environmental Stressors	Revenue Volatility, Market Uncertainty Index, Market Disruption Indicator.
Managerial Dynamics	Budget Allocation Instability.

Construct Category	Variables / Channel Inclusions
Upper-Funnel Media (Broad Awareness)	Meta Spend (Facebook, Instagram), Display Spend, CTV (Connected TV) Spend.
Lower-Funnel Media (High-Intent)	Branded Search Spend, Generic Search Spend, Reddit Spend, LinkedIn Spend, Shopping Spend.
Hybrid/Strategic Media (Dual-Intent)	YouTube Spend, TikTok Spend, OLV (Online Video) Spend, Audio Spend, Affiliate Spend.
Controls	Macro CPI (Inflation), Competitor Index, Revenue Seasonality, Holiday Indicator.
Performance Outcome	Revenue.

Analysis Steps

The analysis followed a systematic structural modeling protocol [13], customized to handle the strict chronological dependencies of time-series data:

Model Specification and Estimation: The causal relationships were mapped to correspond with hypotheses H1–H4, linking environmental stressors to budget instability and sequential media channels to ultimate revenue. The model was implemented using the `semopy` library in Python. The model employed Maximum Likelihood (ML) estimation, wherein the optimizer iteratively sought to minimize the divergence between the observed sample covariance matrix, derived from empirical data, and the model-implied covariance matrix, which was generated by the proposed structural pathways [1].

Model Evaluation and Predictive Validation: Structural fit was assessed using standard indices, targeting a Comparative Fit Index (CFI) ≥ 0.95, a Root Mean Square Error of Approximation (RMSEA) ≤ 0.06, and the coefficient of determination (R^2) [10]. Crucially, because evaluating predictive models on retrospective data risks severe over-fitting, out-of-sample accuracy was validated using a strict chronological 70/30 forward split. The model was trained exclusively on the first 94 weeks and tested on the unseen final 41 weeks to prevent temporal data leakage.

Hypothesis Testing: The structural paths were tested to validate the formulated hypotheses. Path coefficients were examined using a significance level of $\alpha = 0.10$, which is consistent with established practices in marketing research when modeling real-world observational data characterized by high volatility [11].

Benchmark comparison: To assess the strategic value of causal transparency, the SEM framework was compared to a Ridge regression model. This Ridge model represented standard industry "out-of-the-box" Media Mix Models (MMMs). For a direct comparison, the Ridge model was tuned across a range of penalty values, spaced logarithmically from 10^{-3} to 10^{3}, and evaluated using the same 70/30 time-series split. This approach helps determine if the predictive accuracy of a black-box model is enough, or if the causal transparency of SEM provides important strategic insights.

Detailed Research Hypothesis

Based on marketing theory and prior empirical work, four consolidated hypotheses are developed and organized into thematic clusters: (1) managerial behavioral response dynamics, (2) direct revenue attribution mechanisms, (3) mediated funnel mechanics, and (4) strategic resource allocation interdependencies.

To empirically test the theoretical framework without relying on a rigid visual diagram, the Covariance-Based SEM was specified with the following direct paths (→) and covariances (↔):

H1: Managerial Behavioral Response Dynamics. The model specifies direct paths from external market stressors to managerial reactivity: Revenue Volatility, Market Uncertainty Index, and Market Disruption Indicator → Budget Allocation Instability. The resulting impact of this reactive spending is measured via the path Budget Allocation Instability → Revenue, reflecting the assumption that volatility-driven budgeting yields inefficient long-term returns.

H2: Direct Revenue Attribution and Environmental Controls. Direct paths to Revenue are specified for high-intent, lower-funnel channels (Branded Search Spend, Reddit Spend, LinkedIn Spend) and unclarified intent channels (Generic Search Spend). To isolate true advertising incrementality from baseline market fluctuations, environmental controls (Competitor Index, Holiday Indicator, Revenue Seasonality) are also mapped directly to Revenue.

H3: Mediated Funnel Mechanics and Consumer Economics. Sequential funnel effects are modeled by pointing upper-funnel and hybrid platforms toward downstream queries: Meta Spend, Shopping Spend, and YouTube Spend → Branded Search Spend, alongside Display Spend and Shopping Spend → Generic Search Spend.

H4: Strategic Resource Allocation Interdependencies. Managerial budget bundling and substitution behaviors are specified as bidirectional covariances (↔). These capture social and performance interdependencies (Meta Spend ↔ TikTok Spend, LinkedIn Spend, Affiliate Spend, Pinterest Spend), dedicated video bundling (YouTube Spend ↔ TikTok Spend), and broadcast streaming overlaps (CTV Spend ↔ Audio Spend, and OLV Spend ↔ Audio Spend). *To satisfy estimation requirements for these covariances within the computational environment, these hybrid platforms were also mapped as direct controls to the Revenue variable, regardless of individual path significance.*

4 Results

Descriptive Statistics and Baseline Correlations

Prior to structural estimation, a bivariate correlation analysis was performed to examine initial relationships within the 135-week dataset. This analysis indicated considerable multicollinearity among digital interventions, a frequent outcome of algorithmic budget pacing in standard Media Mix Models (MMMs). Specifically, robust positive correlations were identified between upper-funnel and lower-funnel channels, including Meta Spend and Branded Search Spend ($r = 0.548$), and Meta Spend and Generic Search Spend ($r = 0.592$). Moreover, hybrid video platforms demonstrated substantial collinearity, as exemplified by YouTube and OLV Spend ($r = 0.611$).

Additionally, Rolling Revenue Volatility demonstrated a significant positive correlation with Budget Allocation Instability

($r = 0.352$). This dense, inter-correlated matrix underscores the limitations of traditional single-equation regressions; attempting to isolate the independent effect of Meta Spend on Revenue while controlling for Branded Search inevitably suffers from variance inflation. The CB-SEM framework resolves this by explicitly modeling these collinearities as sequential pathways and bidirectional covariances.

Model Fit

The final SEM achieved strong fit indices: CFI = 0.98 (≥ 0.95), RMSEA = 0.03 (≤ 0.06), and $R^2 = 0.82$, indicating the data robustly support the theoretical framework. The model comprises 23 observed variables, yielding 276 unique variances and covariances. With 54 estimated parameters (24 regression paths, 7 covariances, and 23 variances), the model yields 222 degrees of freedom. Because $df > 0$, the model is over-identified, enabling robust parameter estimation and statistical testing. The complete causal architecture and the resulting standardized path coefficients are visually summarized in **Figure 1**. In the figure, standardized path coefficients (β) validate the sequential flow from environmental stressors to managerial action, and from upper-funnel awareness to lower-funnel intent.

Hypothesis Testing Results

The structural pathways within the framework were systematically evaluated to empirically test the four proposed theoretical hypotheses. To determine statistical significance, path coefficients were evaluated at a threshold of $\alpha = 0.10$. While strict laboratory and controlled algorithmic experiments often default to $\alpha = 0.05$, a 90% confidence level is the established and necessary convention in macroeconomic modeling and digital advertising research.

Furthermore, to ensure that structural effects are mathematically comparable across variables with vastly different underlying measurement scales (e.g., aggregate monetary media spend versus percentage-based market uncertainty indices), all reported path estimates are standardized (β). This standardization enables a direct, objective comparison of effect magnitudes, allowing the framework to definitively rank each channel's relative causal influence within the macroscopic recommender system. The comprehensive significance testing results, clustered by their respective behavioral and structural hypotheses, are detailed in **Table 2**.

Indirect Effects

A primary advantage of the macroscopic recommender framework is its ability to mathematically quantify indirect, mediated effects that single-equation models systematically miss. In traditional multiple linear regression or Ridge regression, if an upper-funnel channel does not directly correlate with terminal revenue, its coefficient is shrunk or discarded. CB-SEM resolves this by capturing the hierarchical flow of user intent, calculating indirect causal effects as the product of sequential path coefficients.

Evaluating the sequential paths shows important hidden revenue drivers and funnel bottlenecks that significantly change resource allocation strategies, as shown in **Table 3**:

Meta's Hidden Value (The Support): In standard models, Meta Spend is frequently undervalued because it rarely captures the final, trackable conversion click. However, the SEM reveals a massive, positive indirect effect ($\beta = 0.164$). Meta strongly drives upper-funnel brand awareness ($\beta = 0.733$ to Branded Search), prompting users to subsequently open a search engine, query the brand, and convert ($\beta = 0.224$). Without SEM, an algorithmic optimizer would defund Meta, inadvertently starving the Branded Search channel of its primary traffic source.

YouTube's Substitution Effect: Alternatively, the model indicates a negative indirect effect for YouTube via Branded Search ($\beta = -0.103$). Rather than functioning as an intermediary to search intent, YouTube acts as a substitute within the broader ecosystem. A significant investment in YouTube video interventions is negatively correlated with concurrent Branded Search volume ($\beta = -0.460$), which implies that users who interact with detailed video content either convert directly through video links or delay their search activities, thus cannibalizing concurrent search traffic.

Table 2: Hypothesis Testing Results

Path	Est. (β)	p-value
H1: Managerial Response Dynamics		
Revenue Volatility → Budget Instability	0.436	<0.001
Market Uncertainty → Budget Instability	0.147	0.058
Market Disruption → Budget Instability	-0.210	0.013
Budget Instability → Revenue	-0.025	0.507
H2: Direct Revenue Attribution & Controls		
Branded Search Spend → Revenue	0.224	<0.001
Generic Search Spend → Revenue	-0.081	0.053
Reddit Spend → Revenue	0.205	<0.001
LinkedIn Spend → Revenue	0.103	0.030
Competitor Index → Revenue	0.168	0.001
Holiday Indicator → Revenue	-0.052	0.175
Revenue Seasonality → Revenue	0.557	<0.001
Macro CPI → Revenue	0.412	<0.001
H3: Mediated Funnel Mechanics		
Meta Spend → Branded Search Spend	0.733	<0.001
Shopping Spend → Branded Search Spend	0.349	<0.001
YouTube Spend → Branded Search Spend	-0.460	<0.001
Display Spend → Generic Search Spend	0.475	<0.001
Shopping Spend → Generic Search Spend	0.251	<0.001
H4: Strategic Resource Allocation		
Meta Spend ↔ TikTok Spend	0.577	<0.001
Meta Spend ↔ LinkedIn Spend	0.148	<0.001
Meta Spend ↔ Affiliate Spend	0.238	<0.001
Meta Spend ↔ Pinterest Spend	-0.142	<0.001
YouTube Spend ↔ TikTok Spend	0.462	<0.001
CTV Spend ↔ Audio Spend	0.148	0.001
OLV Spend ↔ Audio Spend	0.161	<0.001

Structural Attrition in Display Media: The framework also successfully identifies structural inefficiencies. Display Spend is very effective at driving a large number of Generic Search queries ($\beta = 0.475$). However, because Generic Search doesn't convert well into Revenue ($\beta = -0.081$), the overall indirect effect of Display on the bottom line is negative ($\beta = -0.038$). This indicates a downstream attrition in the user experiences: Display media successfully generates general product interest, but this interest doesn't have the strong intent needed to drive retail sales.

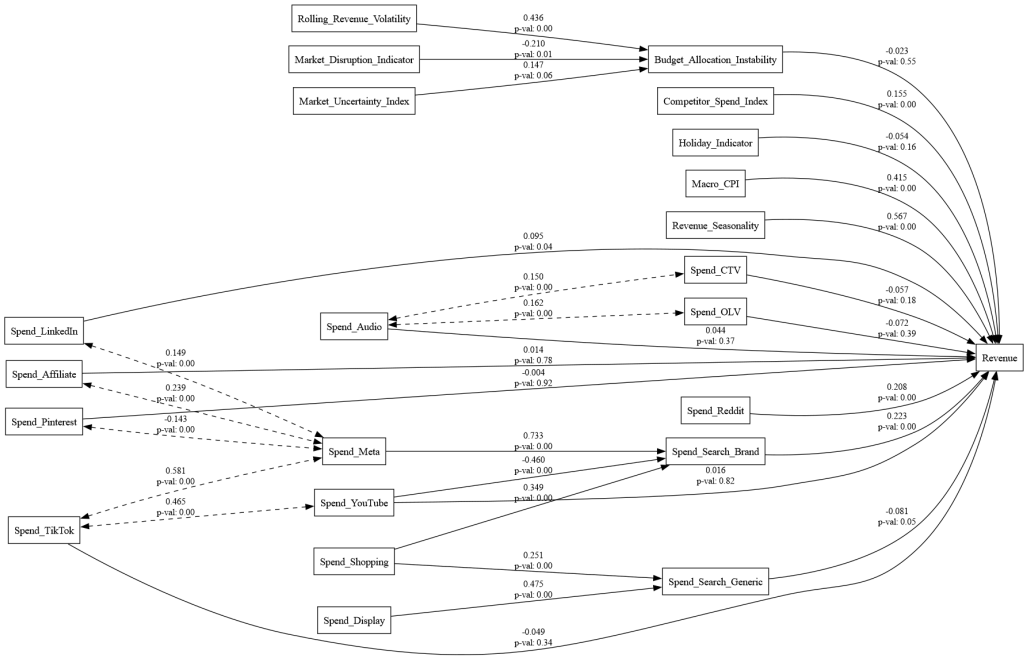

Figure 1: SEM Path Diagram illustrating the causal mechanics of the macroscopic recommender system.

By quantifying these exact behavioral mechanisms, the SEM provides a causal map that prevents the misallocation of resources, proving that optimizing digital media solely on direct, last-touch predictive correlation is mathematically flawed.

Table 3: Indirect Effects and Mediation Analysis

Mediated Path	Calculation	Effect (β)
Meta → Brand Search → Revenue	0.733×0.224	0.164
YouTube → Brand Search → Revenue	-0.460×0.224	-0.103
Display → Non-Brand → Revenue	$0.475 \times (-0.081)$	-0.038

Model Comparison: Predictive Accuracy vs. Causal Transparency

The SEM framework was compared to Ridge Regression (Ridge) and Multiple Linear Regression (MLR) in order to thoroughly confirm the need for the structural approach. Ridge is the algorithmic basis for the, "out-of-the-box" AI attribution models [4], whereas MLR is the conventional baseline. Ridge Regression uses an L2 regularization penalty to address the extreme multicollinearity of marketing data, therefore reducing the coefficients of correlated characteristics to avoid overfitting.

As detailed in **Table 4**, comparing the standardized coefficients across these three frameworks reveals a critical "accuracy paradox." While MLR ($R^2 = 0.858$) and Ridge ($R^2 = 0.854$) achieve slightly higher predictive fit than the SEM ($R^2 = 0.820$), they do so by fundamentally misrepresenting the causal architecture of the user experiences.

Table 4: Comparison of Coefficients Across Models

Variable	MLR	Ridge	SEM
Branded Search Spend	0.051	0.059	0.224
Meta Spend (Indirect)	0.075	0.051	0.164
Reddit Spend	0.142	0.138	0.205
Generic Search Spend	-0.072	-0.068	-0.081
LinkedIn Spend	0.038	0.049	0.103
Budget Instability	0.001	0.003	-0.025

Note: MLR and Ridge use the same predictors as the SEM revenue equation. SEM indirect effect is shown for Meta to enable a direct comparison.

Evaluating the results in **Table 4** highlights three strategic failures inherent in single-equation, black-box models:

- **The Regularization Penalty on Awareness:** Because Meta Spend and Branded Search Spend are highly correlated ($r = 0.548$), the Ridge algorithm interprets their shared variance as redundant noise. Consequently, the L2 penalty aggressively shrinks Meta's coefficient down to a negligible 0.051. SEM corrects this algorithmic blindness by mapping the correlation as a sequential flow, proving that Meta actually carries a massive indirect effect of 0.164.
- **The Undervaluation of Intent:** Both MLR (0.051) and Ridge (0.059) severely undervalue Branded Search compared to the SEM (0.224). Because single-equation regressions force all variables to compete simultaneously for variance in the terminal Revenue node, they fail to isolate the localized, high-intent conversion power of Search.
- **Consistent Independent Drivers:** Conversely, when a digital channel operates as a truly independent, direct driver with lower

cross-channel collinearity, all three models align. Reddit Spend remains mathematically robust across MLR (0.142), Ridge (0.138), and SEM (0.205), proving that traditional regressions are only accurate when the user journey is entirely unmediated.

Finally, traditional models fail to capture the structural drag of managerial decision friction. MLR and Ridge assign a near-zero impact (0.001 and 0.003) to Budget Allocation Instability, whereas SEM properly isolates its negative systemic effect (-0.025).

If a marketing company utilized the Ridge regression as an automated resource recommender, the system would aggressively defund Meta and Branded Search due to their artificially suppressed coefficients. The SEM demonstrates that optimizing a macroscopic recommender system based purely on marginalized predictive error (R^2) actively destroys the structural pathways that generate user intent.

5 Discussion

This study's findings address bridging the explainability gap in marketing analytics. By viewing Media Mix Modeling (MMM) as a recommender system and using Covariance-Based SEM, the research shows that allocating resources based solely on predictive accuracy (R^2) is fundamentally flawed. The results have three main theoretical and strategic implications for creating data-driven allocation frameworks.

The Illusion of Predictive Accuracy

The most important finding of this study is the empirical evidence that good prediction results don't always lead to good strategic insight. As shown in the initial comparisons, opaque models like MLR and Ridge regression successfully reduce errors on new data, achieving a predictive fit similar to the SEM. However, they do this by confusing correlation with causation, which leads to a fundamental misrepresentation of the underlying causes.

Ridge regression, which uses an L2 penalty to handle the multicollinearity often found in sequential user experience, strongly reduces the coefficients of variables that are highly correlated. In practice, this means the algorithm looks at Meta Spend and Branded Search Spend, identifies their shared variance, and effectively zeroes out Meta. If a marketing organization were to deploy this black-box model as an automated budget recommender, the system would advise defunding upper-funnel awareness campaigns. The SEM causal map proves this would be catastrophic: Meta generates a substantial indirect revenue effect ($\beta = 0.164$) by acting as the primary engine that stimulates Branded Search queries. Actionable intelligence requires moving beyond superficial attribution to protect these critical, indirect dependencies.

Deconstructing the Funnel: Synergies, Substitutes, and Attrition

By examining the overall user experience, this study moves beyond basic "return on investment" measures. It highlights the specific roles that different platforms performs within media mix system.

Independent Conversion Engines: The analysis identifies Reddit as a strong, independent revenue driver ($\beta = 0.205$). In contrast to Meta, which requires the intermediary of Branded Search to generate revenue, Reddit directly captures localized, high-intent communities that convert immediate conversions. It was the only paid media channel that demonstrate consistent significance across all predictive and causal models, thereby highlighting its unique position as a direct conversion engine.

Intra-Ecosystem Substitution: Structural equation modeling (SEM) results show that not all upper-funnel interventions work together effectively. YouTube acts as a direct substitute for Branded Search Spend ($\beta = -0.460$). Behaviorally, this suggests that when users are deeply engaged in long-form video content, they either convert through direct embedded links within the video ecosystem itself or delay their external exploratory behavior, actively displacing concurrent search traffic. Recommender models that fail to account for this substitution effect will systematically misallocate budget to video at the expense of search capture.

Diagnosing Structural Attrition: The framework also identifies prime opportunities for strategic budget reallocation by mapping inefficiencies. Generic Search exhibits a negative trend regarding terminal revenue ($\beta = -0.081$). While upstream Display media successfully drives volume into Generic Search, that volume fails to convert. This structural attrition indicates that bidding on unclarified, non-branded search terms yields low-intent traffic, signaling to managers that capital should be reallocated from Generic Search into high-performing direct drivers like Reddit or Branded Search.

The Structural Penalty of Managerial Decision Friction

Finally, this study explicitly models the "human-in-the-loop" dynamics of resource allocation by introducing and validating the metric of Budget Allocation Instability. Traditional algorithmic planners assume that human managers execute budgets perfectly and consistently. However, the data reveals severe decision friction.

When macroeconomic stressors rise—specifically, rolling revenue volatility ($\beta = 0.436$) and market uncertainty ($\beta = 0.147$)—managers react by rapidly shifting budgets week-over-week. Crucially, the structural path from this reactive instability to final Revenue is inherently negative ($\beta = -0.025$). This empirical proof of "decision friction" demonstrates that reactive budget shifting during market volatility yields absolutely no performance benefit; in fact, it acts as a structural drag on the ecosystem.

This finding urges organizations to abandon reactive, week-to-week heuristics. Future algorithmic recommender systems must be designed not just to suggest optimal channel splits, but to actively resist short-term market noise, anchoring human managers to stable, causally validated long-term allocations. Furthermore, the identification of strategic budget bundling—such as the strong covariances between Meta and TikTok (0.577) or CTV and Audio (0.148)—proves that managers buy media in cognitive "bundles." Automated systems must mathematically untangle these psychological bundles to prevent biased recommendations and ensure each platform is evaluated on its isolated causal merit.

6 Conclusion

This research introduces Covariance-Based Structural Equation Modeling (CB-SEM) as a transparent, causally-driven recommendation engine for macroscopic marketing resource allocation. By transitioning away from isolated, single-equation predictive models

like Multiple Linear Regression and Ridge regression, this framework successfully maps the structural interdependencies inherent in complex user experience.

The empirical results prove that optimizing resource allocation based purely on marginalized predictive error (R^2) is strategically inadequate. By quantifying sequential mediation, the SEM framework generates highly actionable budget recommendations—such as properly valuing Meta's massive indirect awareness impact and prioritizing Reddit's unmediated conversion power—moving analytics beyond the flaws of pure predictive correlation. Furthermore, the empirical quantification of managerial "decision friction" provides a data-driven mandate for organizations: algorithmic planners must be designed to anchor human managers to stable, long-term allocations rather than permitting reactive, inefficient budget shifting during periods of market volatility.

This structural framework, while providing a strong strategic basis, serves as the ideal diagnostic engine to guide future experimental design and probabilistic forecasting. As a direct next step, future research will leverage the SEM's structural map to prioritize media channels for targeted Geographic Lift (Geo-Lift) testing. Rather than testing channels blindly, the SEM isolates high-leverage or ambiguous pathways—such as Meta's sequential mediation—allowing data science teams to deploy resource-intensive geo-experiments efficiently. Once these causal effects are empirically validated through Geo-Lift tests, they will be utilized as informative priors within a Bayesian Media Mix Model (MMM). Traditional Bayesian MMMs frequently rely on uninformative priors or black-box algorithmic pacing, which perpetuate the structural undervaluation of upper-funnel media. By injecting these experimentally calibrated, SEM-derived total effects as the expected means for prior distributions, future models will mathematically fuse the causal transparency of structural equations with the dynamic, predictive flexibility of Bayesian inference.

As recommender systems expand beyond micro-level user content curation into macro-level financial resource allocation, explainability is no longer optional. This research demonstrates that structural causal modeling provides the essential transparency required to build organizational trust, correct human bias, and drive true incremental growth in the digital economy.

References

[1] Jichang Bian. 2021. Research on initial trust mechanism of structural equation model based on data . In *2021 2nd International Conference on Computer Science and Management Technology (ICCSMT)*. IEEE Computer Society, Los Alamitos, CA, USA, 27–30. doi:10.1109/ICCSMT54525.2021.00013

[2] Varun Chivukula, Ling Jin, and Julia Zhan. 2026. From Attribution to Causality in Digital Advertising: Blackout Experiments and Incrementality-Adjusted Profitability at Dropbox. *IEEE Access* PP (01 2026), 1–1. doi:10.1109/ACCESS.2026.3670337

[3] Vesna Damnjanovic. 2019. Entry Market Strategy for Weaver Chatbot using the Digital B2B Model. In *2019 International Conference on Artificial Intelligence: Applications and Innovations (IC-AIAI)*. 40–403. doi:10.1109/IC-AIAI48757.2019.00014

[4] Macarena Estevez, María Teresa Ballestar, and Jorge Sainz. 2024. A Primer on Out-of-the-Box AI Marketing Mix Models. *IEEE Transactions on Engineering Management* PP (01 2024), 1–14. doi:10.1109/TEM.2024.3519172

[5] F. V. Ferdinand, A. F. Tanugerah, and K. V. I. Saputra. 2023. Analysis of the Influence of Social Media Marketing on the Purchase Decisions of Consumers Using Structural Equation Modelling (SEM). In *2023 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*. 0495–0499. doi:10.1109/IEEM58616.2023.10406422

[6] Brett Gordon, Florian Zettelmeyer, Neha Bhargava, and Dan Chapsky. 2017. A Comparison of Approaches to Advertising Measurement: Evidence from Big Field Experiments at Facebook. *SSRN Electronic Journal* (01 2017). doi:10.2139/ssrn.3033144

[7] Alexander A. Hernandez, Victor James C. Escolano, Muhammad Syukur, Ace Lagman, Roland A. Calderon, and Rosanna T. Adao. 2024. Predicting the Factors to Artificial Intelligence in Peer-to-Peer Energy Sharing Service Adoption Intention: A Structural Equation Model Assessment. In *2024 9th International Conference on Business and Industrial Research (ICBIR)*. 0841–0846. doi:10.1109/ICBIR61386.2024.10875879

[8] Danping Lin, Yuting Zhang, and Ben Jiang. 2022. Identification of Factors Affecting the Application of Additive Manufacturing Technology in Mass Customization Based on Structural Equation Model. In *2022 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*. 0137–0141. doi:10.1109/IEEM55944.2022.9989796

[9] Javier Marin. 2025. A new framework for Marketing Mix Modeling: Addressing Channel Influence Bias and Cross-Channel Effects. arXiv:2311.05587 [cs.LG] https://arxiv.org/abs/2311.05587

[10] Yikui Mo, Yongning Niu, and Yanbin Fu. 2008. Structural Equation Modeling of Traveler Information Needs. In *2008 International Conference on MultiMedia and Information Technology*. 233–236. doi:10.1109/MMIT.2008.38

[11] I Gusti Ngurah Satria Wijaya, Evi Triandini, and Ezra Tifanie Gabriela Kabnani. 2020. Customer Loyalty in Marketplace: Structural Equation Modeling Analysis. In *2020 2nd International Conference on Cybernetics and Intelligent System (ICORIS)*. 1–5. doi:10.1109/ICORIS50180.2020.9320825

[12] Rita Orji, Julita Vassileva, and Regan Mandryk. 2014. Modeling the Efficacy of Persuasive Strategies for Different Gamer Types in Serious Games for Health. *User Modeling and User-Adapted Interaction* 10.1007/s11257-014-9149-8 (07 2014). doi:10.1007/s11257-014-9149-8

[13] Wei Chin Jess Tan and Hian Chye Koh. 2023. Applying Structural Equation Modelling in Education Research. In *2023 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. 1–8. doi:10.1109/TALE56641.2023.10398315

[14] Tereshchenko, Natala Ushenko, Maryna Dielini, Marja Nesterova, Lozhachevska, and N. Honcharenko. 2021. Behavioral Models of Decision-Making by Business and Industry Stakeholders. *Financial and Credit Activity: Problems of Theory and Practice* 5 (11 2021), 300–313. doi:10.18371/fcaptp.v5i40.245156